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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Autumn 2025

Course Code: CSE-3527 Course Title: Compiler

Full Marks: 30 Time: 1 Hour 30 minutes

[Answer all the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Define a program with a proper block diagram that is used to compile a program in high-level languages and contrast it with the interpreter.	2	CO1	U
	Or			
	Explain the working of a language processing system that converts a high-level source program into an executable target code. Illustrate the process with an appropriate system diagram.	2	CO1	U

b)	Consider the following C program:
	`int main(){`
	`float salary, basic, increment, RATE;`
	`scanf("%f%f%f", basic, increment, RATE);`
	`salary = basic + increment * RATE;`
	`printf("%f\n", salary);`
	`return 0;`
	`}`
	Translate this C program step by step through the compiler phases. Draw the diagram of the compiler's phases with output translating annotations at each phase and show the appropriate symbol table entries for identifiers, syntax definition in context-free grammar (CFG), and tokenization for each lexeme. [Assume all variables are floating type]
		8 (4+1+3)	CO1	Ap

2. a)	What do you mean by syntax definition in compiler construction? Describe each of the basic syntax definition components of a language in compiler design. Give syntax definition for the following conditional if-else statements in C programming:
	`if(condtions){`
	`statements;`
	`} else if(condtions){`
	`statements;`
	`}….`
	`else if(condtions){`
	`statements;`
	`}else{`
	`statements;`
	`}`
		4 (1+1+2)	CO2	Ap

	Or,			
	Describe ambiguity in CFG and its impacts on parser design? Consider the following CFG G_1: $digit \rightarrow digit + digit \mid digit - digit$ $digit \rightarrow digit * digit \mid digit/digit$ $digit \rightarrow 0 \mid \ldots \mid 9$ Whether the above grammar is ambiguous or not. Justify your answer using the derivation or parse tree for the input string "5*2+3-5". If it is ambiguous, convert it into unambiguous grammar using appropriate variables. Construct the syntax tree and left-most derivation for the input string "5*2+3-5".	4 (1+1+2)	CO2	Ap

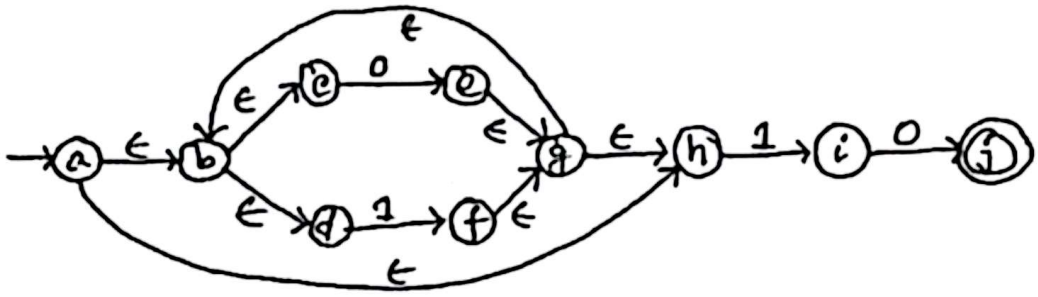
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b) Define a CFG for the language of 0 and 1 to generate the palindrome string. Explain the roles of the lexical analyzer and parser in a compiler with the appropriate block diagram. | 4 (2+2) | CO2

c) Define all four basic types of grammar by emphasizing the production rules of the grammar with appropriate examples. | 2

3. a) Write the regular expression and draw the nondeterministic finite automata (NFA), deterministic finite automata (DFA) transition diagram, ε-NFA and CFG to define a identifier for C programming language (Σ={a-z, A-Z, '_', 0-9}) transition diagram to define the number of the C programming language. | 6 (1+1+1+1.5+1.5) | CO2

b) Consider the following $\epsilon - NFA$ to define the all language (Σ={0,1}) end with one and zero. Convert it into the DFA using the ϵ-closure technique step by steps: | 4 (2+2) | CO2



Or,

Differentiate between DFA and NFA. Describe the language (Σ={a,b}) in and $\epsilon - NFA$ for the regular expression: a(a|b)*a | 4 (1+3) | CO

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)
BSc in CSE, Mid-Term Examination, Spring 2025
Course Code: CSE-3527 Course Title: Compiler
Full Marks: 30 Time: 1 Hour 30 minutes

[Answer all the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Define the compiler with a block diagram that illustrating its front-end and back-end phases. Elucidate the analysis of the source program with its compiling phases.	4 (2+2)	CO1	U
	Or			
	How is an executable target program generated from a given source program? Explain it using a language processing system with an appropriate diagram. What basic tools are needed for compiler construction in a language processing system?	4 (1+2+1)	CO1	An
b)	Translate the following C programming statement step by step through the compiler phases. Draw the diagram of the compiler's phases with output translating annotations at each phase and generate the symbol table for the following statement. `a = x + y * r - w / 4;` [where the values of `a, r, and w` are fractional]	6 (2+3+1)	CO1	An
2. a)	Write is the purpose of context-free grammar (CFG) in compiler design. Define and emphasize the concept of ambiguity in CFG. How does it impact a compiler design? Consider the following CFG G_1: $Exp \rightarrow Exp\ Op\ Exp$ $Exp \rightarrow (Exp)$ $Op \rightarrow +\mid -\mid *\mid /$ $Exp \rightarrow int$ Whether the above grammar is ambiguous or not. Justify your answer using its input string "`int * (int + int)`". Construct the syntax tree for the input string "`int + int * (int / int)`".	6 (3+3)	CO2	Ap
	Or			
	Define synthesized attributes? Consider the following CFG G_2: and G_3 G_2: $S \rightarrow Ab \mid aB$; $A \rightarrow aA \mid \varepsilon$; $B \rightarrow Bb \mid \varepsilon$ G_3: $Seq \rightarrow Seq\ Instr \mid begain$; $Instr \rightarrow east \mid north \mid west \mid south$	6 (1+2+3)	CO2	An
	i) Write the language definition of the above grammar G_2 with $\Sigma = \{a, b\}$ and generate the leftmost and rightmost derivation for the input string *"aaab"*			
	ii) For a robot position tracking system CFG G3, suppose a robot can instructed to move east, north, west, or south from the current position. Position can be marked by a coordinate pair (x, y) with an initial position (0,0) at the beginning. Draw the tracking position of the robot and give the leftmost derivation and parse tree for the input string "`begin west south east east north north`"			

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b) Describe the roles of lexical analyzer and parser in a compiler with the appropriate block diagram. Identify the tokens, lexemes and patterns of the following C programming code segment in a table [please avoid the redundant entry of tokens]: (4, CO2, U) (2+2)

```
int c, n=5, fact = 1;
printf("Enter a integer number\n");
for(c = 1; c <= n; c++)
    fact = fact * c;
printf("Factorial of %d = %d\n", n, fact);
```

3 a) Write the regular expression and draw the nondeterministic finite automata (NFA) transition diagram to define the number of C programming language including fractional and exponential parts. Construct the $\epsilon - NFA$ for regular expression "(0|1)*011" and convert it into the deterministic finite automata (DFA) using ϵ-closure. (6, CO2, An) (2+4)

b) Define the different types of grammar with emphasizing the basic production rules of the grammar. Identify the regular grammars, right-linear grammars, left-linear grammars, context free grammars, context sensitive grammars and useless non-terminals in the following grammars. [Some grammar may fit more than one category]. (4, CO2, Ap) (2+2)

G4:	G5:	G6:	G7:
S → baB	$S \rightarrow aX \mid bY$	$S \rightarrow aA \mid \varepsilon$	$S \rightarrow Ma \mid Mb$
aB → aA	$X \rightarrow aY$	$A \rightarrow Ab \mid a$	$M \rightarrow Nb \mid d$
A → bA|b	$Y \rightarrow b$	$C \rightarrow c$	$N \rightarrow c$

END

L → 0 | . . . | 9

L+ . L*

0.5

0

3

9.5

t4 = id5 / t3

t5 = id4 - t4

t6 = uminus t5

id1 = id2 + t6

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Spring 2024

Course Code: CSE-3527 Course Title: Compiler

Full Marks: 30 Time: 1 Hour 30 minutes

[Answer all of the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Elucidate the block diagram to define a tool to convert an input source program into an output target program and contrast it with the interpreter. What is the significant role of this tool in a language processing system, and illustrate it with an appropriate figure of a language processing system?	4 (2+2)	CO1	U
b)	Translate the following C programming assignment statement step by step in compiler phases. Draw the diagram of the compiler's phases with output annotation of each phase and symbols table. `net_salary = basic + house_rent - basic * tax_rate;` [where the value of `tax_rate` is fractional]	6 (4+2)	CO1	An
2. a)	Illustrate the concept of ambiguity and how it impacts on a compiler design? Consider the following CFG G_1: $Exp \rightarrow Exp + Exp \mid Exp - Exp$ $Exp \rightarrow Exp * Exp \mid Exp / Exp$ $Exp \rightarrow 0\mid1\mid2\mid3\mid4\mid5\mid6\mid7\mid8\mid9$ $Exp \rightarrow id$ Justify whether the above grammar is ambiguous using its input string "6-5+2*1/id". If it is ambiguous, convert it into the unambiguous grammar and construct the syntax tree for the same input string.	5 (2+3)	CO2	Ap
	Or			
	What do you mean by synthesized attributes? Consider the following CFG G_2: $E \rightarrow E + T \mid E - T$ $E \rightarrow T$ $T \rightarrow 0\mid1\mid2\mid3\mid4\mid5\mid6\mid7\mid8\mid9$ i) Write the syntax directed definition using semantic rules for the above CFG G_2 and Create an annotated parse tree for the input string: *"9-5+2-1"* ii) Generate leftmost and rightmost derivation for the input string: *"5+1-4+8"*	5 (1+2+2)	CO2	An
b)	Illustrate the roles of lexical analyzer in a compiler with schematic diagram. What is the main advantage of separating the analysis phase of compiling into lexical analysis and parsing? Identify the tokens, lexemes and patterns of the following C programming function in a table:	5 (2+1+2)	CO2	U

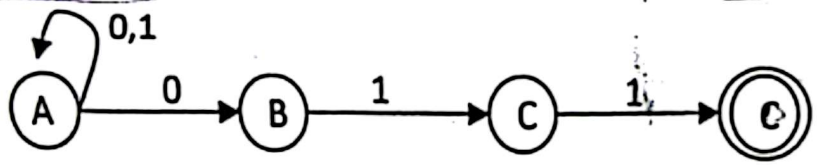
```
long factorial (int n) {
    if (n==1)
        return 1;
    else
        return n*factorial(n-1);
}
```


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3. a) Why finite automata is important in a compiler construction? Write the regular expression to define the identifier of C programming language and construct the $\epsilon - NFA$ and DFA using that regular expression. | 5 (1+1+2+1) | CO2 | An

Or

Contrast DFA with NFA. Construct the equivalent NFA for the regular expression RE_1: $(a|b)^*abb(a|b)$ directly and Convert the following NFA to DFA. | 5 (1+2+2) | CO2 | An



b) Define syntax directed translation of *if-else* conditional statement of C programming language using appropriate grammar. What is the main factor to identify the types of grammar? Identify the regular grammars, right-linear grammars, left-linear grammars, context free grammars, context sensitive grammars and useless non-terminals in the following grammars. [Some grammar may fit more than one category]. | 5 (2+3) | CO2 | Ap

G3:	G4:	G5:	G6:
$S \rightarrow aM$	$S \rightarrow aX \mid \varepsilon$	$A \rightarrow aA \mid \varepsilon$	$S \rightarrow Aa \mid AbTy \mid$
$M \rightarrow Pb$	$X \rightarrow aY$	$B \rightarrow Ab \mid b$	$B \rightarrow Cb \mid Dd$
$Ma \rightarrow bPc$	$Y \rightarrow b$	$C \rightarrow c$	$C \rightarrow c$
$Pc \rightarrow ac$			

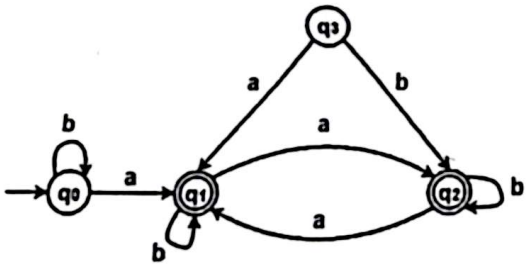
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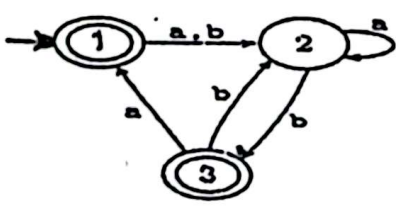
International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Autumn 2023
Course Code: CSE-3527 Course Title: Compiler
Total marks: 30 Time: 90 minutes

1.	a.	What is bytecode? How do Java language processors work?	2	CO1	C2
	b.	How is an executable target program generated? Explain it using a language processing system.	3	CO1	C2
	c.	Translate the following assignment statement step by step using compiler phases: $z = a * b + c/4$; [$z, a, b,$ and c contain floating point values]	4	CO1	C3
2.	a.	Consider the CFG: $S \rightarrow +SS \mid *SS \mid S \mid a$	2	CO1	C1
		1. Show that: aa^+a^* can be generated by the grammar.			C2
		2. Draw the parse tree			
	b.	Consider the grammar: $P \rightarrow aPbP \mid bPaP \mid \in$ Consider $ababab$ as the string to check the ambiguity of the given grammar.	4	CO1	C3
	c.	Construct the regular expressions and provide two accepted strings for each. (Any two):	4	CO2	C3
		1) $\{w \mid w$: starts with a 'b' and has at least one a$\}$			
		2) $\{w \mid w$: does not contain the substring abab$\}$			
		3) $\{w \mid w$: set of strings consisting of even number of a's followed by b$\}$			
		4) $\{w \mid w$: set of strings of 1's and 0's in even length$\}$			
3.	a.	Minimize the following DFA:	4	CO2	C3



	b.	Convert the following regular expression directly to its equivalent NFA: $abb\ (a\|b) * ba$ OR, $(a\|b)^*a(a\|b)(a\|b)$	2	CO2	C3
	c.	Convert the following NFA to DFA:	4	CO1	C3



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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Autumn 2022

Course Code: CSE 3527 Course Title: Compiler

Total marks: 30

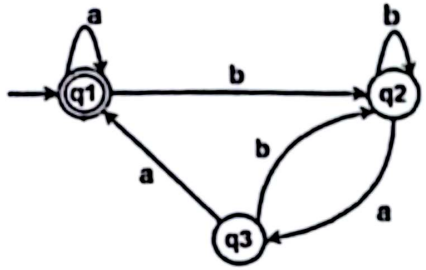
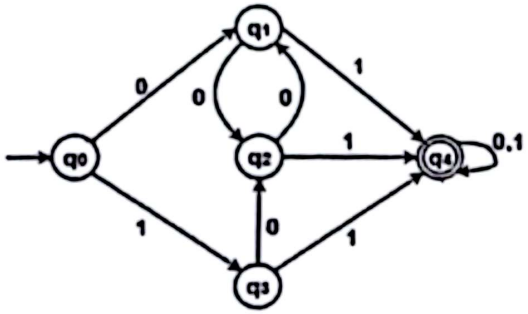
Time: 1 Hour and 30 minutes

			CO	DL
1.				
a)	Translate the following assignment statement step by step using compiler phases: z = c/d + a*b, here z, c, d are integer and a, b are floats.	5	CO1	C3

b)	Write the differences between lexical analyzer and parser. Identify the lexemes and their corresponding tokens and non-tokens from the following C code:
	#include<stdio.h>
	#define a 10
	int sum(float x, float y)
	{
	float m;
	// this will add 2 numbers
	sum= x+y;
	if(x>y)
	printf("%d", sum);
	else
	return sum;
	}
		5	CO1	C2, C3

2.				
a)	What is context sensitive grammar? Identify the useless non terminals, Left Linear Grammar, Right Linear Grammar from the following grammar? S -> ABd/a A -> eBC/b B -> aB/C C -> aC/B Or, When do we call a grammar ambiguous? Check whether the given grammar G is ambiguous or not for the string "a(a)aa". A → AA A → (A) A → a	4	CO3	C1, C2
b)	Construct CFG for the language of all non-Palindromes. Or, Construct a CFG for the language L = 0^n1^{4n} where n>=1.	2	CO3	C3
c)	Write the regular expressions for the following languages: i.All strings of a's and b's with an even number of a's and an odd number of b's. ii.All strings of a's and b's that do not contain the substring abb.	4	CO2	C3
3.				
a)	Find the regular expression for the following FA:	4	CO2	C3

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a)	Minimize the following DFA:  Or, Convert the regular expression R= abb(a|b)*(a|b) directly to its equivalent NFA	4	CO2	C3
b)	Describe the languages denoted by the following regular expressions: i. a(a|b)*a ii. a*ba*ba*ba*	2	CO2	C3

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Exam, 5th Semester, Spring 2022

Course Code: CSE 3527 **Course Title: Compiler**

Total marks: 30 **Time: 1 hours 30 minutes**

[Answer all the questions; in some questions, there are options; Figures in the right-hand margin indicate full marks.]

1. a. Specify each phaseof a compiler explicitly with corresponding input and output while processing the following statement: **5**

Expression := Term * factor + 128

b. Consider the grammar: **5**

$$
\begin{aligned}
stmt &\rightarrow if\text{-}stmt \mid \textbf{other} \\
if\text{-}stmt &\rightarrow \textbf{if (} exp \textbf{)} stmt \\
&\mid \textbf{if (} exp \textbf{)} stmt \textbf{ else } stmt \\
exp &\rightarrow \textbf{0} \mid \textbf{1}
\end{aligned}
$$

i. Now, show that the grammar is ambiguous by constructing two parse trees for the following string:
if (0) if (1) other else other

ii. Rewrite the grammar so that it becomes unambiguous.

2. a. Consider the following grammar: **3**

E→ E or T | T
T →T and F | F
F→ not F | (Σ) | true | false

Now, construct a parse tree for the following sequence:

not (true or false)

b. Suppose you are considering DFA from regular expression. Now, compute the *followpositions* for the following regular expression: **7**
(a|b)*ab*ab

OR

Convert the regular expression R=(1|0)*1 to its equivalent NFA using Thompson construction.

3. a. i. What are the advantages of separating the analysis phase of compiling into lexical analysis and parsing? **5**

ii. Why context free grammar is preferred over regular expressions in parsing?

b. Convert the regular expression R=(a|b)*abb(a|b) directly to its equivalent DFA. **5**

OR

Prove that 0*(10*)* is equivalent to (1*0)*1*

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